

Document 6: The Audio Routing Matrix Specification

6.1 Compact Signal Chain Arrays

The real-time Audio I/O Service manages a tightly packed linear array of node definitions containing the virtual plugin implementation instance pointer and the parent slot index link. Unpopulated or bypassed slots are completely omitted from this tracking block, avoiding loops over empty slots or unnecessary buffer copies.

6.2 Dual-Topology Crossfade Transition Architecture

When a background compilation finishes and alters the routing sequence, the engine runs both the old and new signal paths concurrently during a temporary transition window to prevent audio artifacts.

- **Crossfade Duration:** Fixed at exactly 1,024 audio samples (~23.2ms at 44.1kHz).
- **Double-Buffer Execution:** For the duration of the 1,024-sample window, incoming audio blocks are fed to both the Current Audio Chain and the new Next Audio Chain simultaneously.
- **Interpolation Logic:** Output sample arrays are generated by applying linear gain modifiers across the block, smoothly crossfading from the old configuration to the new one:

$$Output[n] = (Stream_{Current}[n] \times (1.0 - g[n])) + (Stream_{Next}[n] \times g[n])$$

Where $g[n]$ represents the linear gain ramping smoothly from 0.0 to 1.0 across the 1,024-sample boundary. Once the counter reaches 1024, the old chain is safely cleared out, the new chain becomes primary, and double-processing terminates.